

SentioScape: Facilitating Effective Cross-Cultural Communication through Emotional Mixed Reality

Jack Brophy* Dunya Donna Chen† Tanner Person‡ Kouta Minamizawa§ Giulia Barbareschi¶

Keio University
Graduate School of Media Design

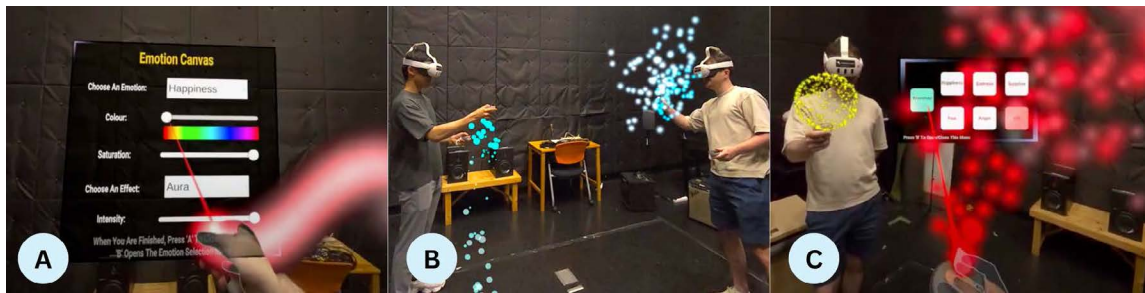


Figure 1: (A) Customization: A user creates their own visual representations of their emotions using SentioScape's 'Emotion Canvas' menu. This particular visualization shows a red aura. (B) Communication: Two users communicate while using the SentioScape system. (C) Comprehension: One user activates the 'Translate' feature to better understand the other user's visualizations

Three images showing the key stages of the SentioScape experience.

ABSTRACT

Increasing global migration has led to the rapid growth of interactions between organizations, groups, and individuals from various cultures. As such, knowing how to decode communication involving conversational partners from other cultures is essential in avoiding misunderstandings and needless conflicts. We present SentioScape, a Mixed Reality (MR) system for aiding emotional expression and understanding to foster more effective cross-cultural communication. The system allows co-located users to customize their own abstract visual representations of their emotions and perceive those of another person. 20 participants took part in an experiment to evaluate the system, with preliminary results revealing some positive trends that suggest the system has potential as a tool for creating more effective emotional communication between high- and low-context communicators.

Index Terms: XR, mixed reality, cross-cultural communication, emotion visualization.

1 INTRODUCTION

According to the International Migration Organization, over 280 million people are currently living outside their country of nationality. Increasing global migration means that the necessity of intercultural competence is no longer limited to expatriates, but a fundamental requirement for an increasingly diverse society. To address this challenge, our research draws on existing communication models such as the Shannon-Weaver model, which illustrates the process of information transmission from a sender to a receiver

[1]. We seek to create a system that streamlines the communication process by reducing noise and aiding users throughout the encoding and decoding process of emotions during active exchanges, thus enabling more effective communication. Our research focuses on communication between people of low- and high-context cultures. The former are described as cultures where emotional valence in communication is more direct and explicit, whereas the latter are cultures where emotions are expressed in communication more subtly and implicitly according to the situation. The differences between these two approaches affect the process of encoding and decoding during both verbal and non-verbal exchanges, which can lead to misunderstandings [3]. A culture cannot be characterized as exclusively high context or low context, though some cultures tend to be at the lower end while others are at the higher end of the continuum [5].

In the last few decades, technology has had a tremendous impact on how we communicate with each other, both synchronously and asynchronously. Thanks to the increased diffusion of XR headsets and building on the popularity of social VR platforms, social XR is believed to offer opportunities for supporting new modalities of communication between individuals in a variety of settings [6]. Substantial research has investigated the use of social XR between remotely located individuals. Work exploring the use of XR systems that provide contextual emotional information during real-time communication and aid emotional expression and understanding has also been gaining increased attention. An example is Heightened Empathy, a multi-user interactive XR experience that places users in immersive representations of each other's emotional state [2]. This system aims to help people collaborate and understand each other by automatically creating visualizations based on affective states measured using physiological sensors. While such an approach requires no direct input, it relies on the approximation of emotional states, which do not necessarily provide sufficient support for more complex communication. Our study aims to address this research gap through the development and evaluation of SentioScape, a mixed reality system for aiding emotional expression and understanding during real-time communication.

*e-mail: jbrophy@kmd.keio.ac.jp

†e-mail: dchen@kmd.keio.ac.jp

‡e-mail: tannerperson@kmd.keio.ac.jp

§e-mail: kouta@kmd.keio.ac.jp

¶e-mail: barbareschi@kmd.keio.ac.jp

2 SENTIOSCAPE

The design process of the SenticScape system began with an ideation workshop to explore how different people would like to visualize their emotions in XR, and in what kind of scenarios they would like to use such a visualization system. The SenticScape system was developed based on the outcomes of this workshop and subsequently refined to improve usability in two separate iterations. SenticScape was developed for the Meta Quest 3¹ using Unity 3D². Multiplayer functionality was implemented using the Normcore multiplayer framework³. SenticScape enables users to create simple customized visualizations for five basic emotions, which can be triggered when interacting with others to reinforce verbal and non-verbal messages to foster more effective cross-cultural communication between co-located users. The system also features a 'Translate' function, which enables individuals to view the other person's emotion visualization according to their own created schema, thus helping understanding by providing automatic decoding.

3 METHODOLOGY

The final version of SenticScape was evaluated through a mixed-method study with 20 participants from an international graduate school. Eleven participants identified their gender as "Woman", while nine identified as "Man". Four participants were between the ages of 18-24, fifteen were aged between 25-34, and one was aged between 34-45.

3.1 High/Low Context Communication Scale

Before the evaluation session, participants completed a survey to determine their High/Low context communication score. The High/Low context communication scale used to categorize the participants was originally developed by Ohashi (2000) [4]. For the purposes of the evaluation, the participants were paired together and split into two groups. Group A was made up of those who placed on opposite sides of the high/low context communication scale. Group B consisted of people who scored similarly on the scale.

3.2 Procedure

In pairs, the participants were asked to complete three rounds of activities that were designed to encourage interactions in three scenarios: non-verbal communication, expressing agreement/disagreement, and engaging in emotive situations. Each round of activities featured a different condition: The first was face-to-face communication with no technology at all (No MR), the second was using the SenticScape system with the 'Translate' feature disabled (MR), the third was using the SenticScape system with the 'translate' feature enabled (MR w/ Translation). The order of the rounds was randomized in each session. Participants completed a questionnaire after each round of activities. The questionnaires were used to gather quantitative data, while qualitative data was collected from an interview conducted at the end of the session. The Repeated Measures ANOVA test was used to determine statistically significant differences in the data. All interviews were transcribed and thematic analysis was performed to draw conclusions.

3.3 Key Findings

Results from the experiments show that the system can improve users' perception of their ability to understand the emotions of another person, especially during non-verbal communication scenarios. Crucial to enhanced understanding is the ability to translate another's visualizations. Moreover, the positive impact of SenticScape was more pronounced for participants in Group A. Statistical analysis did not show a significant difference between the MR

conditions. However, there were positive trends of increased understanding when utilizing MR with translation feature compared to other conditions. For the non-verbal activity, there was an increase in the percentage of people who agreed or strongly agreed that they could understand the emotions of another person between the No MR (40%), MR (50%), and MR w/ Translation (60%) rounds. There was a similar trend following the agreement/disagreement activity (85% MR w/ Translation, 75% No MR, 65% MR), as well as the emotive situations activity (80% MR w/ Translation, both 70% No MR and MR). Statistical analyses was also carried out to investigated specific trends for both groups. For Group A, the positive impact of the MR w/ Translation round becomes more evident with a mean difference from No MR that ranges between 0.3 points and 0.6 points. However, for Group B, the mean scores for both the MR and MR w/ Translation rounds are almost always lower than for the No MR round. This suggests that the system may be beneficial when used by two people who have opposing High/Low context communication scores, but amongst those who have similar communication styles, the system may actually become an obstacle. This indicates that when communication barriers are low the system creates unnecessary cognitive load for individuals and can also hide important information, such as facial expression, due to the presence of the HMD. On the other hand, when communication styles differ and the risk of miscommunication higher, SenticScape offers an important tool for transmitting contextual information.

4 CONCLUSION & FUTURE WORK

SenticScape aims to facilitate opportunities for user collaboration and participation in the process of interpersonal communication. Moreover, it seeks to enhance the expression of users' emotional states through visual means, with an aim to find novel solutions to challenges that arise in cross-cultural communication scenarios. The experiments conducted to evaluate the system revealed some positive trends that suggest SenticScape has potential as a tool for creating more effective emotional communication between high- and low-context communicators. In future works we hope to test the system in group communication scenarios and incorporate physiological sensing for further insights into the effectiveness of the system.

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¹<https://www.meta.com/jp/en/quest/quest-3/>

²<https://unity.com/>

³<https://normcore.io/>